

George Emil Sadek

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Professional Summary

AI/ML Engineer candidate and final-year Computer Engineering student with hands-on experience building LLM applications, RAG workflows, prompt-engineered assistants, backend APIs, and Dockerized environments. Skilled in Python, FastAPI, Flask, LangChain, LlamaIndex, CrewAI, SQL, and Linux, with practical experience in retrieval pipelines, tool-using agents, document ingestion, API integration, and technical communication.

Core Skills

Programming & AI: Python, SQL, JavaScript, C++, scikit-learn, PyTorch, TensorFlow, Pandas, NumPy
LLM & Agentic Systems: LLM applications, RAG, prompt engineering, LangChain, LlamaIndex, CrewAI, tool use, multi-step agent workflows, embeddings, semantic retrieval, response evaluation
Backend & Integration: FastAPI, Flask, REST APIs, JSON services, API consumption, SDK-style integration, request/response handling, backend logic design
Data Engineering: Data preprocessing, ETL-style pipelines, document ingestion, metadata structuring, dataset preparation, SQL querying
Cloud & DevOps: Docker, Git, Linux, CI/CD concepts, secure coding practices, containerized workflows, AWS ML Foundations; familiarity with Vertex AI and AWS Bedrock ecosystems
Databases: MySQL, SQLite, SQL Server

Relevant Projects

Student Helper – RAG-Based Learning Assistant Personal Project

- Built a retrieval-augmented assistant that uses user-provided study materials to generate grounded answers instead of generic LLM output.
- Designed the pipeline for document ingestion, chunking, embeddings, semantic retrieval, prompt construction, and answer generation.
- Structured document metadata and retrieval flow to improve context quality, answer accuracy, and traceability across user queries.

Rehab Helper – Medical AI Assistant and Web Platform Graduation Project

- Contributing to a scalable AI-enabled medical platform that combines predictive models with a knowledge-grounded assistant for patient support.
- Helped design safe-response boundaries, user workflows, backend structure, and AI feature integration for a maintainable multi-model system.
- Collaborated on product and engineering decisions around therapist messaging, progress tracking, and documentation of workflow behavior.

Agentic Workflow Prototyping Applied AI Engineering

- Prototyped agent-based workflows using CrewAI, LangChain, and LlamaIndex for task decomposition, retrieval, tool use, and routing.
- Improved reliability by debugging generated outputs, tightening prompts, and restructuring workflows into clearer components.
- Studied LangGraph-style orchestration patterns and stateful agent design principles for controllable production-minded workflows.

Email Spam Detection Machine Learning Project

- Developed an NLP classification pipeline using Python and scikit-learn with preprocessing, TF-IDF feature engineering, model training, and evaluation.

Education & Training

B.Sc. in Computer Engineering Sep 2021 – Jun 2026
Faculty of Engineering at Shoubra, Benha University CGPA: 3.4/4.0

Training: AI Training – NTI | Data Analysis Training – NTI | Entrepreneurship Training – ITIDA / Plan International
Certifications: Machine Learning Specialization (Stanford / Andrew Ng), Deep Learning Specialization, AWS ML Foundations

Leadership, Awards & Additional Information

Shell Eco-marathon Autonomous Programming Competition – 2nd Place with Shoubra Racing Team; contributed to autonomous systems work involving simulation, perception, debugging, and engineering collaboration.

AI Leader, IEEE Shoubra Team – Supported peers in AI and machine learning activities and helped communicate technical concepts clearly.

Languages: Arabic (Native), English (Professional) | **Work Style:** Open to full-time AI/ML engineering roles; comfortable collaborating with technical and non-technical stakeholders and documenting technical decisions clearly